

IB MATH AA SL

## Lesson 19

# Trigonometric Identities

Trigonometry · 60-minute lesson

Syllabus SL 3.6 · Property of Lynda Badmus Education

$\theta$

# Do Now — recall

## ON YOUR WHITEBOARD

1. Write  $\tan \theta$  in terms of  $\sin$  and  $\cos$ .
2. What is  $\sin^2 \theta + \cos^2 \theta$  ?
3. Simplify  $\sin \theta / \cos \theta \times \cos \theta$ .

## Answers (click to reveal)

1.  $\tan \theta = \sin \theta / \cos \theta$
2. 1
3.  $\sin \theta$

# What we're learning today

## LEARNING OBJECTIVES

- 1 State**  
the Pythagorean identity.
- 2 Use**  
the quotient identity for tan.
- 3 Apply**  
the double-angle identities.
- 4 Simplify**  
and prove trig expressions.

## SUCCESS CRITERIA — I CAN...

- ✓  
use  $\sin^2\theta + \cos^2\theta = 1$
- ✓  
swap between  $\tan \theta$  and  $\sin/\cos$
- ✓  
apply the double-angle formulas
- ✓  
simplify a trig expression to a single ratio

# Core identities

## YOU MUST KNOW

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

### Pythagorean

comes from the unit circle  $x^2 + y^2 = 1$

### Rearrange

$\sin^2 \theta = 1 - \cos^2 \theta$ , and vice-versa

## DOUBLE ANGLE

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

also

$$\cos 2\theta = 1 - 2\sin^2 \theta = 2\cos^2 \theta - 1$$

# Use the identity

## QUESTION

Given  $\sin \theta = 3/5$  and  $\theta$  is acute, find  $\cos \theta$ .

## SOLUTION

Use  $\sin^2 \theta + \cos^2 \theta = 1$ :

$$\cos^2 \theta = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25}$$

So  $\cos^2 \theta = 16/25$ , and  $\theta$  acute  $\rightarrow$  positive:

$$\cos \theta = \frac{4}{5}$$

# Your turn — identities

## ANSWER IN YOUR BOOK

1. If  $\cos \theta = 4/5$  (acute), find  $\sin \theta$ .
2. Simplify  $\sin^2 \theta + \cos^2 \theta + 1$ .
3. Write  $1 - \cos^2 \theta$  more simply.

## Answers (click to reveal)

1.  $3/5$
2.  $2$
3.  $\sin^2 \theta$

# Double-angle formulas

## DOUBLING $\theta$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

### Use

to rewrite  $\sin 2\theta$  or  $\cos 2\theta$  in terms of  $\theta$

### Choose

the  $\cos 2\theta$  form that suits the problem

## THREE FORMS OF $\cos 2\theta$

$$\cos^2 \theta - \sin^2 \theta$$

$$1 - 2 \sin^2 \theta$$

$$2 \cos^2 \theta - 1$$

# Double angle

## QUESTION

Given  $\sin \theta = \frac{3}{5}$  and  $\cos \theta = \frac{4}{5}$ ,  
find  $\sin 2\theta$ .

## SOLUTION

Use  $\sin 2\theta = 2 \sin \theta \cos \theta$ :

$$\sin 2\theta = 2 \times \frac{3}{5} \times \frac{4}{5}$$

Multiply out:

$$= \frac{2 \times 3 \times 4}{5}$$

# Exam-style question

Given  $\cos \theta = 5/13$  and  $\theta$  is acute.

- (a) Find  $\sin \theta$ . [2]
- (b) Hence find  $\sin 2\theta$ . [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a)  $\sin^2 \theta = 1 - 25/169 = 144/169$  (M1)  $\rightarrow \sin \theta = 12/13$  (A1)
- (b)  $\sin 2\theta = 2 \cdot (12/13) \cdot (5/13)$  (M1)(A1)  $= 120/169$  (A1)

# Common mistakes



## Squaring notation

$\sin^2\theta$  means  $(\sin \theta)^2$ , not  $\sin(\theta^2)$ .



## Sign of the root

Decide + or – using the quadrant before taking the square root.



## Double angle $\neq 2\times$

$\sin 2\theta \neq 2 \sin \theta$ .



## Pythagorean rearrange

$\sin^2\theta = 1 - \cos^2\theta$  (not  $1 + \cos^2\theta$ ).

# Mixed practice

## QUESTIONS

1. Simplify  $1 - \sin^2\theta$ .
2. If  $\sin\theta = 5/13$  (acute), find  $\cos\theta$ .
3. Write  $\sin 2\theta$  if  $\sin\theta=1/2$ ,  $\cos\theta=\sqrt{3}/2$ .
4. Simplify  $\sin\theta/\cos\theta$ .

## Answers (click to reveal)

1.  $\cos^2\theta$
2.  $12/13$
3.  $2 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$
4.  $\tan\theta$

# Plenary & exit ticket

## KEY FORMULAS

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

## EXIT TICKET — before you leave

1. Simplify  $1 - \cos^2 \theta$ .
2. Write  $\tan \theta$  in terms of  $\sin$ ,  $\cos$ .
3. State  $\cos 2\theta$  (one form).

### Answers (click to reveal)

1.  $\sin^2 \theta$
2.  $\sin \theta / \cos \theta$
3.  $\cos^2 \theta - \sin^2 \theta$

# Homework

## SET ON DR FROST MATHS (auto-marked)

### Task:

Trigonometric identities.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

## NEXT LESSON

### Lesson 20

Solving trigonometric equations over a given interval.

$$\sin \theta = k$$